Deep Generative Models

Lecture 13

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Recap of Previous Lecture

Let us choose $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$. Then $p(\mathbf{z}) = p(\mathbf{x}_0, \mathbf{x}_1) = p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)$.

$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)\right); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

Let's consider straight conditional paths:

$$\boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0 \quad \boldsymbol{\sigma}_t(\mathbf{x}_1) = \epsilon$$

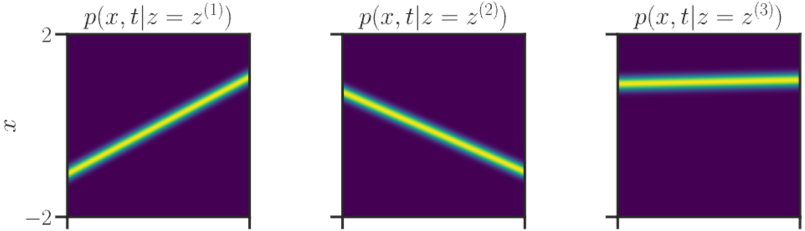


image credit: A Visual Dive into Conditional Flow Matching

Recap of Previous Lecture

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\theta}(\mathbf{x}(t),t)-\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$

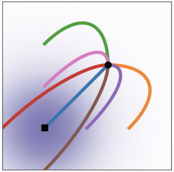
$$p_t(\mathbf{x}|\mathbf{x}_1) = q_{1-t}(\mathbf{x}|\mathbf{x}_0 = \mathbf{x}_1)$$

Variance Exploding SDE

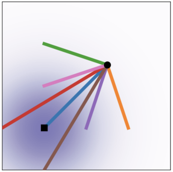
$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\mathbf{x}_1, \sigma_{1-t}^2\mathbf{I}\right) \quad \Rightarrow \quad \mathbf{v}(\mathbf{x},\mathbf{x}_1,t) = -\frac{\sigma_{1-t}'}{\sigma_{1-t}}(\mathbf{x}_t - \mathbf{x}_1)$$

Variance Preserving SDE

$$p_t(\mathbf{x}_t|\mathbf{x}_1) = \mathcal{N}\left(\alpha_{1-t}\mathbf{x}_1, (1-\alpha_{1-t}^2)\mathbf{I}\right) \Rightarrow \mathbf{v}(\mathbf{x}_t,\mathbf{x}_1,t) = \frac{\alpha_{1-t}'}{1-\alpha_{1-t}^2}\cdot(\alpha_{1-t}\mathbf{x}_t - \mathbf{x}_1)$$



Diffusion



OT

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Recap of Previous Lecture

Continuous state space

- **Discrete time** $t \in \{0, 1, \dots, T\}$ $\Rightarrow$ **DDPM / NCSN**.
- **Continuous time** $t \in [0, 1]$ $\Rightarrow$ **Score-based SDE models**.

Discrete state space

- **Discrete time** $t \in \{0, 1, \dots, T\}$.
- **Continuous time** $t \in [0, 1]$.

Key advantages of discrete diffusion

- Parallel generation
- Flexible infilling
- Robustness
- Unified framework

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https://aaronlou.com/blog/2024/discrete-diffusion/

Recap of Previous Lecture

Discrete Diffusion Markov Chain

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \text{Cat}(\mathbf{Q}_t \mathbf{x}_{t-1}),$$

Each $\mathbf{x}_t \in \{0,1\}^K$ is a **one-hot vector** encoding the categorical state (it is just one token).

Transition Matrix

$$[\mathbf{Q}_t]_{ij} = q(x_t = i | x_{t-1} = j), \qquad \sum_{i=1}^{K} [\mathbf{Q}_t]_{ij} = 1.$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \text{Cat}(\mathbf{Q}_{1:t} \mathbf{x}_0), \qquad \mathbf{Q}_{1:t} = \mathbf{Q}_t \mathbf{Q}_{t-1} \cdots \mathbf{Q}_1.$$

- The choice of $\mathbf{Q}_t$ determines how information is erased and what the stationary distribution becomes.
- $\mathbf{Q}_t$ and $\mathbf{Q}_{1:t}$ should be easy to compute for each t.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Recap of Previous Lecture

Uniform vs. Absorbing Transition Matrix

Aspect	**Uniform Diffusion**	**Absorbing Diffusion**
$\mathbf{Q}_t$	$(1-\beta_t)\mathbf{I} + \beta_t\mathbf{U}$	$(1-\beta_t)\mathbf{I} + \beta_t\mathbf{e}_m\mathbf{1}^\top$
$\mathbf{Q}_{1:t}$	$\bar{\alpha}_t\mathbf{I} + (1-\bar{\alpha}_t)\mathbf{U}$	$\bar{\alpha}_t\mathbf{I} + (1-\bar{\alpha}_t)\mathbf{e}_m\mathbf{1}^\top$
$\mathbf{Q}_{1:\infty}$	$\mathbf{U}$	$\mathrm{Cat}(\mathbf{e}_m)$
Interpretation	Random replacement	Gradual masking of tokens
Application	Image diffusion	Text diffusion $\approx$ Masked LM

Observation

Both schemes gradually destroy information, but differ in their stationary limit. Absorbing diffusion bridges diffusion and masked-language-model objectives.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Recap of Previous Lecture

Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right).$$

ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big).$$

- The reverse process $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}\big(\boldsymbol{\pi}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\big)$ is a learned categorical distribution.
- Minimizing $\mathcal{L}_t$ w.r.t. $\boldsymbol{\theta}$ is equivalent to minimizing the cross-entropy

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \mathrm{H}\Big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0),\, p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\Big).$$

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Recap of Previous Lecture

Independent Token-wise Forward Process

$$q(\mathbf{X}_t|\mathbf{X}_{t-1}) = \prod_{i=1}^{m} q(\mathbf{x}_t^i|\mathbf{x}_{t-1}^i) = \text{Cat}(\mathbf{Q}_t\mathbf{X}_{t-1})$$

$$p_{\boldsymbol{\theta}}(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

- All distributions defined by the forward process are factorized.
- Each factor conditions on the entire noisy sequence $\mathbf{X}_t$ — this is exactly the **masked language modeling** pattern.

Final objective: masked LM

$$\mathcal{L} = \sum_{t=1}^{T}\sum_{i=1}^{m} \mathbb{E}_{q(\mathbf{X}_t|\mathbf{X}_0)}\Big[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_t)\Big].$$

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Outline

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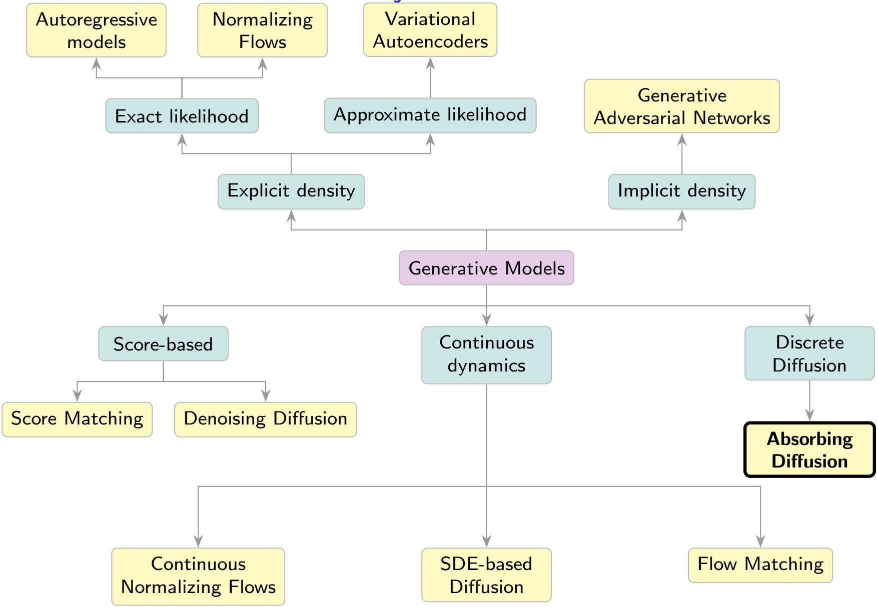
Generative Models Taxonomy



Absorbing Diffusion: Forward Process

Let's restrict to the case of absorbing transition matrix.

$$\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t\, \mathbf{e}_m \mathbf{1}^\top, \qquad \bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s).$$

$$\mathbf{Q}_{1:t} = \bar{\alpha}_t\, \mathbf{I} + (1-\bar{\alpha}_t)\, \mathbf{e}_m \mathbf{1}^\top.$$

Sahoo S. et al. Simple and effective masked diffusion language models, 2024

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$$\mathbf{Q}_{1:t} = \bar{\alpha}_t\,\mathbf{I} + (1-\bar{\alpha}_t)\,\mathbf{e}_m\mathbf{1}^\top.$$

Each position is either still clean or already masked:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \bar{\alpha}_t[\mathbf{x}_t = \mathbf{x}_0] + (1-\bar{\alpha}_t)[\mathbf{x}_t = \mathbf{e}_m]$$

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$$\mathbf{Q}_t = \begin{pmatrix} 1-\beta_t & 0 & 0 \\ 0 & 1-\beta_t & 0 \\ \beta_t & \beta_t & 1 \end{pmatrix} \quad \Rightarrow \quad \text{the masked state is absorbing.}$$

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What happens in the conditioned reverse process $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$?

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Absorbing Diffusion

Conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} [\mathbf{x}_{t-1} = \mathbf{x}_t], & \text{if } \mathbf{x}_t \neq \mathbf{e}_m, \\ \rho_t[\mathbf{x}_{t-1} = \mathbf{x}_0] + (1-\rho_t)[\mathbf{x}_{t-1} = \mathbf{e}_m], & \text{if } \mathbf{x}_t = \mathbf{e}_m, \end{cases}$$

where

$$\rho_t = \frac{\beta_t\, \bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}.$$

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Absorbing Diffusion

Conditioned reverse distribution

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where

$$\rho_t = \frac{\beta_t\,\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}.$$

- If $\mathbf{x}_t \neq \mathbf{e}_m$, then the token must be unchanged: $\mathbf{x}_{t-1} = \mathbf{x}_t$.
- Observing an unmasked token at time t fixes the entire history: $\mathbf{x}_{t-1} = \mathbf{x}_t = \cdots = \mathbf{x}_0$.

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- If $\mathbf{x}_t \neq \mathbf{e}_m$, then the token must be unchanged: $\mathbf{x}_{t-1} = \mathbf{x}_t$.
- Observing an unmasked token at time t fixes the entire history: $\mathbf{x}_{t-1} = \mathbf{x}_t = \cdots = \mathbf{x}_0$.
- If $\mathbf{x}_t = \mathbf{e}_m$, the previous token may be either clean or masked.
- With probability ρ_t, masking occurred exactly at step t (so $\mathbf{x}_{t-1} = \mathbf{x}_0$).
- With probability $(1-\rho_t)$, the token was already masked earlier (so $\mathbf{x}_{t-1} = \mathbf{e}_m$).

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Absorbing Diffusion

Sequence Distribution

$$q(\mathbf{X}_{t-1}|\mathbf{X}_t, \mathbf{X}_0) = \prod_{i=1}^{m} q(\mathbf{x}_{t-1}^i|\mathbf{x}_t^i, \mathbf{x}_0^i).$$

Each position i has two possible cases:

$$\mathbf{x}_t^i \neq \mathbf{e}_m \ \Rightarrow \ \mathbf{x}_{t-1}^i = \mathbf{x}_t^i, \qquad \mathbf{x}_t^i = \mathbf{e}_m \ \Rightarrow \ \mathbf{x}_{t-1}^i \in \{\mathbf{x}_0^i, \mathbf{e}_m\}.$$

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Interpretation

- The forward process produces **random partial observations** of the clean sequence.
- If a token is visible at time t, the reverse distribution is deterministic.
- Unmasked tokens yield a deterministic posterior and therefore contribute only a constant to the ELBO. Therefore, only masked tokens contribute to the training loss.

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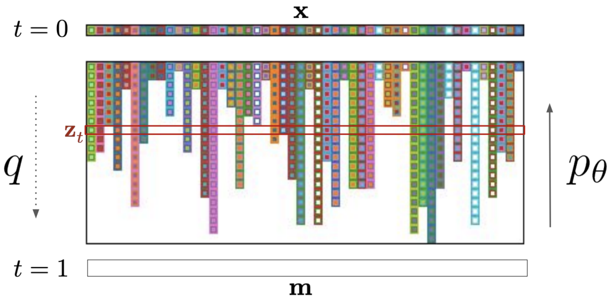
Absorbing Diffusion



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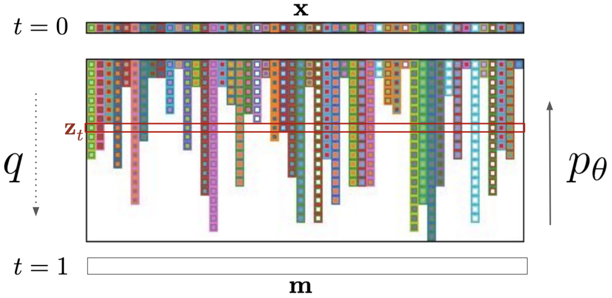
Absorbing Diffusion



$$p_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^{m} p_\theta(\mathbf{x}^i_{t-1}|\mathbf{X}_t).$$

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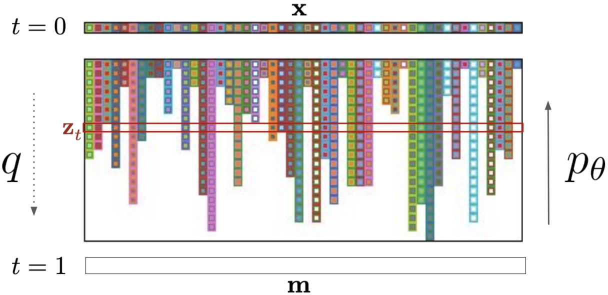
Absorbing Diffusion



$$p_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^{m} p_\theta(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

Objective: sequence-level $\mathcal{L}_t$

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{X}_t|\mathbf{X}_0)} \sum_{i=1}^{m} \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_\theta(\mathbf{x}_0^i|\mathbf{X}_t) \right] + \text{const}$$

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Algorithm: Discrete-Time Absorbing Diffusion

Training

1. Sample $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X}),\ t \sim U\{1, \ldots, T\}$.
2. Corrupt the sequence by independent masking:
$$\mathbf{x}_t^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } 1 - \bar{\alpha}_t, \\ \mathbf{x}_0^i, & \text{with prob. } \bar{\alpha}_t, \end{cases} \qquad i = 1, \ldots, m.$$
3. Predict token distributions in parallel:
$$p_{\boldsymbol{\theta}}(\mathbf{X}_0|\mathbf{X}_t) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_t).$$
4. Compute masked-CE loss:
$$\mathcal{L}(\boldsymbol{\theta}) = \rho_t \sum_{i=1}^{m} [\mathbf{x}_t^i = \mathbf{e}_m] \big[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_t) \big].$$

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Algorithm: Discrete-Time Absorbing Diffusion

Sampling

1. Initialize $\mathbf{X}_T \leftarrow \mathbf{e}_m \mathbf{1}^\top$ (fully masked).
2. For $t = T, T-1, \ldots, 1$:
   2.1 Predict $p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_t)$ for all positions.
   2.2 For each masked position ($\mathbf{x}_t^i = \mathbf{e}_m$):
   $$\mathbf{x}_{t-1}^i = \begin{cases} \hat{\mathbf{x}}_0^i \sim p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_t), & \text{with prob. } \rho_t, \\ \mathbf{e}_m, & \text{with prob. } 1 - \rho_t. \end{cases}$$
   2.3 For each unmasked position: $\mathbf{x}_{t-1}^i = \mathbf{x}_t^i$.
3. Return the final sequence $\mathbf{X}_0$ (fully unmasked).

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Outline

From Discrete Time to Mask Rate

In absorbing diffusion, the forward process is

$$q(\mathbf{x}_t|\mathbf{x}_0) = \bar{\alpha}_t[\mathbf{x}_t = \mathbf{x}_0] + (1 - \bar{\alpha}_t)[\mathbf{x}_t = \mathbf{e}_m].$$

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From Discrete Time to Mask Rate

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- The distribution depends on t only through the scalar

$$\lambda_t = 1 - \bar{\alpha}_t \in [0, 1].$$

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- We can therefore reparameterize the corruption level by

$$t \quad \Rightarrow \quad \lambda \in [0, 1].$$

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- We can therefore reparameterize the corruption level by

$$t \quad \Rightarrow \quad \lambda \in [0, 1].$$

- We directly define a family of corrupted distributions indexed by a continuous mask rate λ:

$$q(\mathbf{x}_\lambda|\mathbf{x}_0) = (1 - \lambda)[\mathbf{x}_\lambda = \mathbf{x}_0] + \lambda[\mathbf{x}_\lambda = \mathbf{e}_m].$$

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Discrete ELBO Revisited

Recall the per-step ELBO term for absorbing diffusion:

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{X}_t|\mathbf{X}_0)} \sum_{i=1}^{m} \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[- \log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_t)\right] + \text{const.}$$

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Discrete ELBO Revisited

Recall the per-step ELBO term for absorbing diffusion:

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \sum_{i=1}^{m} \rho_t [\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_t) \right] + \text{const.}$$

Replacing the discrete index t with the continuous mask rate λ, the training objective becomes

$$\mathcal{L} = \int_0^1 w(\lambda) \mathbb{E}_{q_\lambda(\mathbf{x}_\lambda|\mathbf{x}_0)} \sum_{i=1}^{m} [\mathbf{x}_\lambda^i = \mathbf{e}_m] \left[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_\lambda) \right] d\lambda.$$

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Recall the per-step ELBO term for absorbing diffusion:

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Interpretation

Training corresponds to optimizing a **continuous mixture of masked language modeling objectives** with different mask rates.

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Algorithm: Masked Diffusion Language Model (MDLM)

Training

1. Sample $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X})$, $\lambda \sim U[0, 1]$.
2. Corrupt the sequence by independent masking:
$$\mathbf{x}_\lambda{}^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } \lambda, \\ \mathbf{x}_0^i, & \text{with prob. } 1 - \lambda, \end{cases} \qquad i = 1, \dots, m.$$
3. Predict token distributions in parallel:
$$p_{\boldsymbol{\theta}}(\mathbf{X}_0 | \mathbf{X}_\lambda) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_\lambda).$$
4. Compute masked-CE loss:
$$\mathcal{L}(\boldsymbol{\theta}) = w(\lambda) \sum_{i=1}^{m} [\mathbf{x}_\lambda{}^i = \mathbf{e}_m] \big[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_\lambda) \big].$$

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Algorithm: Masked Diffusion Language Model (MDLM)

Sampling

1. Initialize $\mathbf{X}_{\lambda_1} \leftarrow \mathbf{e}_m \mathbf{1}^\top$ (fully masked).
2. For a decreasing schedule $1 = \lambda_1 > \lambda_2 > \cdots > \lambda_L = 0$, $\ell = 1, \ldots, L-1$:
   2.1 Predict $p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_{\lambda_\ell})$ for all positions.
   2.2 For each masked position ($\mathbf{x}_{\lambda_\ell}^i = \mathbf{e}_m$):
   $$\mathbf{x}_{\lambda_{\ell+1}}^i = \begin{cases} \hat{\mathbf{x}}_0^i \sim p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_{\lambda_\ell}), & \text{with prob. } 1 - \frac{\lambda_{\ell+1}}{\lambda_\ell}, \\ \mathbf{e}_m, & \text{with prob. } \frac{\lambda_{\ell+1}}{\lambda_\ell}. \end{cases}$$
   2.3 For each unmasked position: $\mathbf{x}_{\lambda_{\ell+1}}^i = \mathbf{x}_{\lambda_\ell}^i$.
3. Return the final sequence $\mathbf{X}_{\lambda_L}$ (fully unmasked).

Note: Once unmasked, a token stays unmasked. So for a currently masked token:

$$\mathbb{P}(\mathbf{x}_{\lambda_{\ell+1}}^i = \mathbf{e}_m \mid \mathbf{x}_{\lambda_\ell}^i = \mathbf{e}_m) = \frac{\lambda_{\ell+1}}{\lambda_\ell}.$$

Sahoo S. et al. Simple and effective masked diffusion language models, 2024

Outline

Course Overview: Problem Statement

Goal

Learn a generative model $p_{\boldsymbol{\theta}}(\mathbf{x})$ that matches the data distribution $p_{\text{data}}(\mathbf{x})$.

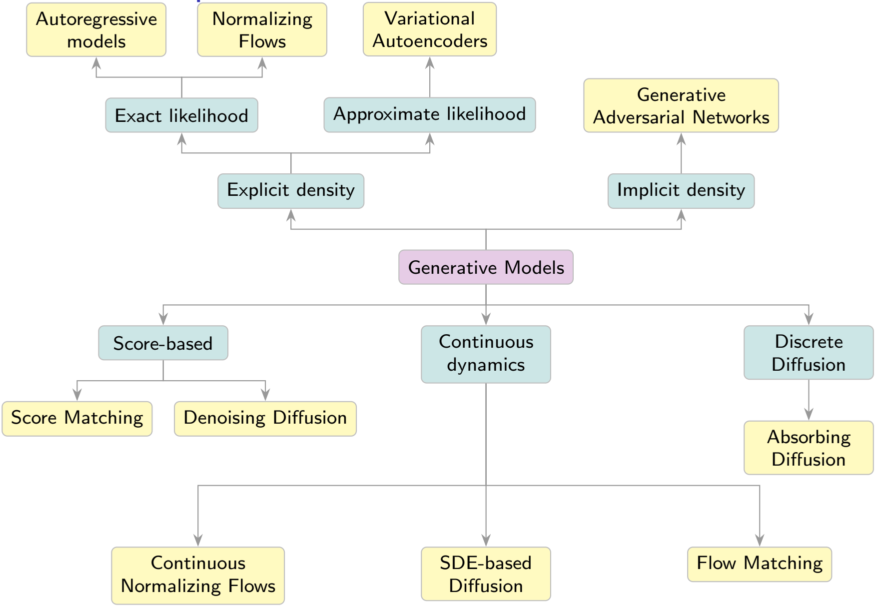
Three lenses on the same problem

- **Divergence minimization** (L1, L5):

$$\min_{\boldsymbol{\theta}} D\left(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}\right) \qquad (\text{KL, JS, Wasserstein, } \ldots)$$

- **Likelihood-based** (L1–L4, L7–L8): maximize $\log p_{\boldsymbol{\theta}}(\mathbf{x})$ (AR, NF, VAE, diffusion as hierarchical VAE).
- **Score-based** (L6–L10): learn $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ (SM / NCSN, DDPM, score-based SDE).

Course Roadmap



Likelihood-based: Exact & Approximate Density (L1–L4)

Autoregressive (L1)

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(x_j|\mathbf{x}_{1:j-1}) \qquad \text{exact likelihood, sequential sampling}$$

Normalizing Flows (L2)

$$\mathbf{x} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{z}), \quad \log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log p(\mathbf{z}) + \log \left| \det \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right|$$

VAE / VQ-VAE (L3–L4)

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) \geq \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}\big(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\big)$$

- Amortized inference + reparametrization trick.
- ELBO surgery $\Rightarrow$ optimal prior; VQ-VAE with discrete latents.

Implicit & Score-based Foundations (L5–L6)

GAN / WGAN (L5)

- Implicit $p_{\boldsymbol{\theta}}$, adversarial min-max

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \mathbb{E}_{p_{\text{data}}} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log\big(1 - D_{\boldsymbol{\phi}}(G_{\boldsymbol{\theta}}(\mathbf{z}))\big)$$

- Optimal discriminator $\Rightarrow$ JSD; WGAN uses Wasserstein distance via Kantorovich–Rubinstein duality.
- Evaluation tools: FID, Precision/Recall, CLIP score, human eval.

Score Matching / NCSN (L6)

- Learn $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ via denoising SM.
- Sample with (annealed) Langevin dynamics over multiple noise scales.

Diffusion Models (L7–L8)

Gaussian diffusion (L7)

- Forward $q(\mathbf{x}_t|\mathbf{x}_{t-1})$, direct sampling $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$.
- Diffusion as hierarchical VAE; ELBO splits into reconstruction + prior KL + $\sum_t \mathcal{L}_t$.

DDPM (L8)

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2, \qquad \boldsymbol{\epsilon}_{\boldsymbol{\theta}} \ \propto \ -\sigma_t \mathbf{s}_{\boldsymbol{\theta}}$$

- Equivalent to score-based generative modeling (NCSN-style).

Guidance (L8)

$$\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{y}) = (1-\gamma)\,\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \varnothing) + \gamma\,\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{y})$$

Continuous Dynamics & SDE (L9–L10)

Neural ODE / continuous-time NF (L9)

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t), \qquad \frac{d}{dt}\log p_t(\mathbf{x}(t)) = -\operatorname{tr}\left(\frac{\partial \mathbf{v}_{\boldsymbol{\theta}}}{\partial \mathbf{x}}\right)$$

Forward SDE (L9–L10)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$$

- KFP equation; VE / VP SDEs unify NCSN and DDPM.

Reverse SDE & Probability flow ODE (L10)

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g^2(t)\, \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right] dt + g(t)\, d\bar{\mathbf{w}}$$

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) - \tfrac{1}{2} g^2(t)\, \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t) \quad \text{(deterministic equivalent)}$$

Flow Matching (L10–L11)

Flow Matching objective (L10)

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,\, p_t(\mathbf{x})} \left\| \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, t) \right\|^2$$

Target $\mathbf{v}(\mathbf{x}, t)$ is the marginal vector field — intractable in general.

Conditional Flow Matching (L11)

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{t,\, p(\mathbf{z}),\, p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \right\|^2$$

- $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathrm{FM}} = \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathrm{CFM}}$ — regress against *conditional* field.
- Conditioning on endpoint $\mathbf{z} = \mathbf{x}_1$ (one-sided) or pair $(\mathbf{x}_0, \mathbf{x}_1)$ (two-sided).
- Straight Gaussian paths $\Rightarrow$ closed-form $\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \mathbf{x}_1 - \mathbf{x}_0$.

Discrete Diffusion (L12–L13)

Forward discrete process (L12)

Categorical $\mathbf{x}_t$ with transition matrices $\mathbf{Q}_t$:

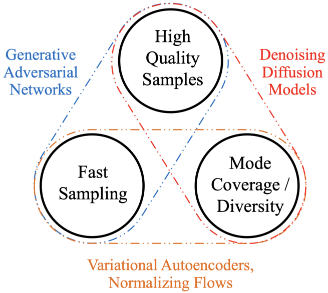
- **Uniform**: $\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t \mathbf{U}$
- **Absorbing**: $\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t\, \mathbf{e}_m \mathbf{1}^\top$ (mask token)

Reverse process & Discrete ELBO (L12)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathrm{Cat}\left(\frac{\mathbf{Q}_t \mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right),$$

each $\mathcal{L}_t$ reduces to a **cross-entropy** loss against $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$.

Generative Learning Trilemma



Rule of thumb

- **Likelihood + Coverage ⇒ AR / NF**
  exact density, *slow sampling*.
- **Likelihood + Fast Sampling ⇒ VAE**
  tractable bound, *blurry samples*.
- **Quality + Fast Sampling ⇒ GAN**
  sharp, *mode collapse, no likelihood*.
- **Quality + Coverage ⇒ Diffusion**
  stable training, *slow sampling*.
- **Quality + Faster Sampling ⇒ FM / ODE**
  fewer steps, *approx. likelihood*.
- **Discrete + Coverage ⇒ Discrete diffusion**
  parallel denoising, *iterative decoding*.

Xiao Z., Kreis K., Vahdat A. Tackling the generative learning trilemma with denoising diffusion GANs, 2021

Summary

- In the absorbing case only masked tokens contribute to the ELBO, which reduces to a continuous mixture of MLM losses over the mask rate $\lambda \in [0, 1]$.
- MDLM sampling performs iterative parallel refinement from fully masked to fully unmasked sequences.
- No generative model dominates all axes of the trilemma — each family trades off likelihood, sample quality, mode coverage and sampling speed.
- Modern frontier (flow matching, discrete diffusion, distilled diffusion) tries to push the Pareto frontier of this trilemma.